THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS FULL SYLLABUS PAPER SET 04

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A point lies on the x-axis. Its y-coordinate is: 1

(a) equal to its x-coordinate **(b)** negative

(c) 0 **(d)** 1

2. The sequence $7, 12, 17, 22, \ldots$ forms a linear pattern. Its nth term is: 1

(a) $5n + 7$ **(b)** $7n + 5$ **(c)** $5n - 12$ **(d)** $5n + 2$

3. The number $0.\overline{6}$ in the form $\frac{p}{q}$ is: 1

(a) 1 **(b)** $\frac{3}{2}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{23}{30}$

4. Expand $(5x + 2)^2$: 1

(a) $25x^2 + 4$ **(b)** $25x^2 + 10x + 4$ **(c)** $25x^2 + 20x - 4$ **(d)** $25x^2 + 20x + 4$

5. Factorise $x^2 - x - 42$: 1

(a) $(x - 6)(x + 7)$ **(b)** $(x - 7)(x - 6)$ **(c)** $(x - 7)(x + 6)$ **(d)** $(x + 6)(x + 7)$

6. An arc subtends an angle of 100° at the centre. The angle it subtends at a point on the remaining part of the circle is: 1

(a) 50° **(b)** 200° **(c)** 100° **(d)** 130°

7. Taking $\pi = \frac{22}{7}$, the circumference of a circle of radius 3.5 cm is: 1

(a) 11 cm **(b)** 38.5 cm **(c)** 44 cm **(d)** 22 cm

8. A sample shows a probability of $\frac{3}{10}$ that a student likes cricket. In a school of 2000 students the expected number who like cricket is: 1

(a) 700 **(b)** 800 **(c)** 300 **(d)** 600

9. The sum of the first 20 natural numbers is: 1

(a) 210 **(b)** 209 **(c)** 420 **(d)** 211

10. A number that is the square of a prime has exactly: 1

(a) 4 factors **(b)** 2 factors

(c) 3 factors **(d)** an even number of factors

11. **Assertion (A):** If a proposition is true, its converse must also be true. 1
Reason (R): The proposition "If a number is a multiple of 6, then it is a multiple of 3" is true, but 9 is a multiple of 3 and not of 6.
(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

12. **Assertion (A):** If equal quantities of 10% and 20% jaggery lemonade are mixed, the result has 15% jaggery. 1
Reason (R): With equal quantities, the weighted average becomes the simple average of 10% and 20%.
(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

13. Marks of 82, 74, 90 are combined with weights 1 : 2 : 2. The weighted average is: 1
(a) 83 **(b)** 81 **(c)** 82 **(d)** 84

14. The list of common divisors of 60 and 24, in increasing order, ends with: 1
(a) 6 **(b)** 60 **(c)** 12 **(d)** 13

15. The 'long division' method of finding the gcd using Indian place-value numerals is found in the work of: 1
(a) Euclid **(b)** Archimedes **(c)** Āryabhaṭa **(d)** Al-Khwārizmī

16. For tiling the plane, the angles meeting at a point of the tiling must add up to: 1
(a) 180° **(b)** 90° **(c)** 360° **(d)** 720°

17. A line with a positive slope: 1
(a) falls as we move from left to right **(b)** is horizontal
(c) rises as we move from left to right **(d)** is vertical

18. The solution of the pair 3x + y = 11 and 2x − y = -1 is: 1
(a) (-2, 5) **(b)** (2, -5) **(c)** (5, 2) **(d)** (2, 5)

19. The volume of a cube is 27 cm^3. Its total surface area is: 1
(a) 9 cm^2 **(b)** 54 cm^2 **(c)** 36 cm^2 **(d)** 27 cm^2

20. The volume of a cone is _____ the volume of a cylinder with the same base and the same height. 1
(a) twice **(b)** half of **(c)** one-third of **(d)** one-fourth of

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 20–30 words each.

21. Show that the points $A(1,\ 1)$, $B(3,\ 3)$ and $C(5,\ 5)$ are collinear using distances. 2

22. The sum of two numbers is 64 and one is 10 more than the other. Form a linear equation and find the numbers. 2

23. In quadrilateral ABCD, $\angle A = 100^\circ$ and $\angle C = 80^\circ$. Is it cyclic? Give a reason. 2

24. A card is drawn from a well-shuffled pack of 52 cards. Find the probability that it is 2
(i) a king
(ii) a red king.

25. Give a real-life example of a proposition that is true but whose converse is false. 2

26. What is the difference between a stacked bar chart and a 100% stacked bar chart? 2

27. Why must the angles meeting at a vertex of a tiling of the plane add up to $360°$? 2

28. Find the points where the line $3x + 4y = 12$ cuts the axes. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each.

29. **Case study:** A triangular plot has medians of lengths 36 m, 27 m and 45 m. A water tank is placed at the centroid. 3
(i) Find the distance of the centroid from the vertex on the 36 m median.
(ii) Find its distance from the midpoint of that side.
(iii) State the theorem that gives these.

30. A ball of radius 7 cm is to be painted. Find the area to be painted and the cost at ₹0.25 per cm^2 ($\pi = \frac{22}{7}$). 3

31. 2 adult and 3 child tickets cost ₹1050; 3 adult and 2 child tickets cost ₹1200. Find the cost of one adult ticket and one child ticket. 3

32. ABCD is cyclic, $\angle BAC = 30°$ and $\angle BCD = 100°$. Find $\angle CBD$. 3

33. A circular ring has outer radius 14 cm and inner radius 7 cm. Find its area ($\pi = \frac{22}{7}$). 3

34. Triangle ADM has vertices $A(3, 4)$, $D(7, 1)$, $M(9, 6)$. It is reflected in the y-axis. Write the image vertices and show that $A'D' = AD$. 3

35. A square has side $(2x + 3)$ units. Find its area and perimeter. 3

OR

Factorise $6x^2 + 7x + 2$ by splitting the middle term. 3

36. A bar chart shows two bars: Family A spends ₹7000 with 20% on healthcare; Family B spends ₹4000 with 30% on healthcare. Which family spends more rupees on healthcare? 3

OR

Explain why a stacked bar chart makes it harder to read the exact value of a middle segment than a cluster-column chart. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. (i) A shop mixes 5 kg of tea at ₹200 per kg with 3 kg of tea at ₹280 per kg. Find the price per kg of the blend. 5
(ii) At what price should the blend be sold to earn 20% profit?

OR

Case study: A salt lake has water of 25% salinity and a river has water of 0.5% salinity. 5
(i) Find the salinity when 1 litre of lake water is mixed with 4 litres of river water.
(ii) Can a mixture with 3% salinity be made? Find the quantity of river water per litre of lake water.
(iii) Can a mixture with 30% salinity be made? Why?

38. **Case study:** A triangular park ABC has sides AB $= 60$ m, BC $= 80$ m, CA $= 100$ m. Paths are laid joining the midpoints D, E, F of AB, BC, CA. 5
(i) Find the lengths of the three paths.
(ii) How many triangles are formed and what can you say about them?
(iii) Find the total length of paths.
(iv) Is $\triangle ABC$ right-angled? Give a reason.

OR

Case study: A craftsman's linkage forms parallelogram ABCD with AB $= 36$ cm, BC $= 20$ cm and $\angle ABC = 65^\circ$. E and F are the midpoints of AB and CD. 5
(i) Find $\angle BCD$ and $\angle ADC$.
(ii) Show that AECF is a parallelogram.
(iii) Find AE and CF.
(iv) Find the perimeter of the linkage.

39. (i) List the integer-sided rectangles of perimeter 40 units and find their areas. 5
(ii) Which has the greatest area and which the least?
(iii) What do you notice about the shape of the rectangle of greatest area?

OR

(i) Prove that the two paths from P to Q — one a semicircle and the other made of n semicircles on adjacent parts of PQ — have equal length. 5
(ii) Hence find the length of each path if $PQ = 28$ cm and the smaller semicircles have diameters 7, 7, 14 cm. ($\pi = \frac{22}{7}$)

40. **Case study:** A cube has edge $(a + b)$. 5
(i) Split it into 2 cubes and 6 cuboids and state their volumes.
(ii) Write the identity.
(iii) If $a = 2$, $b = 3$, verify the identity numerically.

OR

Case study: A rectangular garden has length $(2x + 3)$ m and breadth $(3x + 1)$ m. 5
(i) Write and simplify its area.
(ii) Find the area when $x = 4$.
(iii) A square flower bed has side $(x + 2)$; expand its area and compare with a $(x + 3) \times (x + 1)$ plot.

END OF QUESTION PAPER **Total: 80 Marks**

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